Central Limit Theorem

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What is the central limit theorem?

To understand the central limit theorem (CLT) consider the following exercise. Every day I come to UCLA and measure the heights of a random subset of the student population. I find 50 students, ask their heights, and record the data. Each day I compute the sample average of the 50 measurements, computed as:

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i = \frac{1}{50}\sum_{i=1}^{50} X_i$$

The CLT says that if I do this experiment repeatedly over many days, my list of sample averages will approximately follow a normal distribution. Written mathematically:

$$\bar{X}_n \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Here μ and σ^2 are the population mean and variance. This result holds no matter what the distribution of UCLA student heights is. We can use software, such as R, to simulate the CLT using a variety of probability distributions.

Simulation

In the code below I use R's built-in random number generators to produce 4 populations with different distributions: uniform, exponential, normal, and binomial. I set each population size at 500,000.

```r
set.seed(33) # Fix R's random number generator for reproducible results

pop_size    <- 500000
sample_size <- 50
num_samples <- 5000

population_unif <- runif(pop_size)
population_exp  <- rexp(pop_size, rate = 2)
population_norm <- rnorm(pop_size, mean = 5, sd = 3)
population_bin  <- rbinom(pop_size, size = 5, prob = 0.3)
```

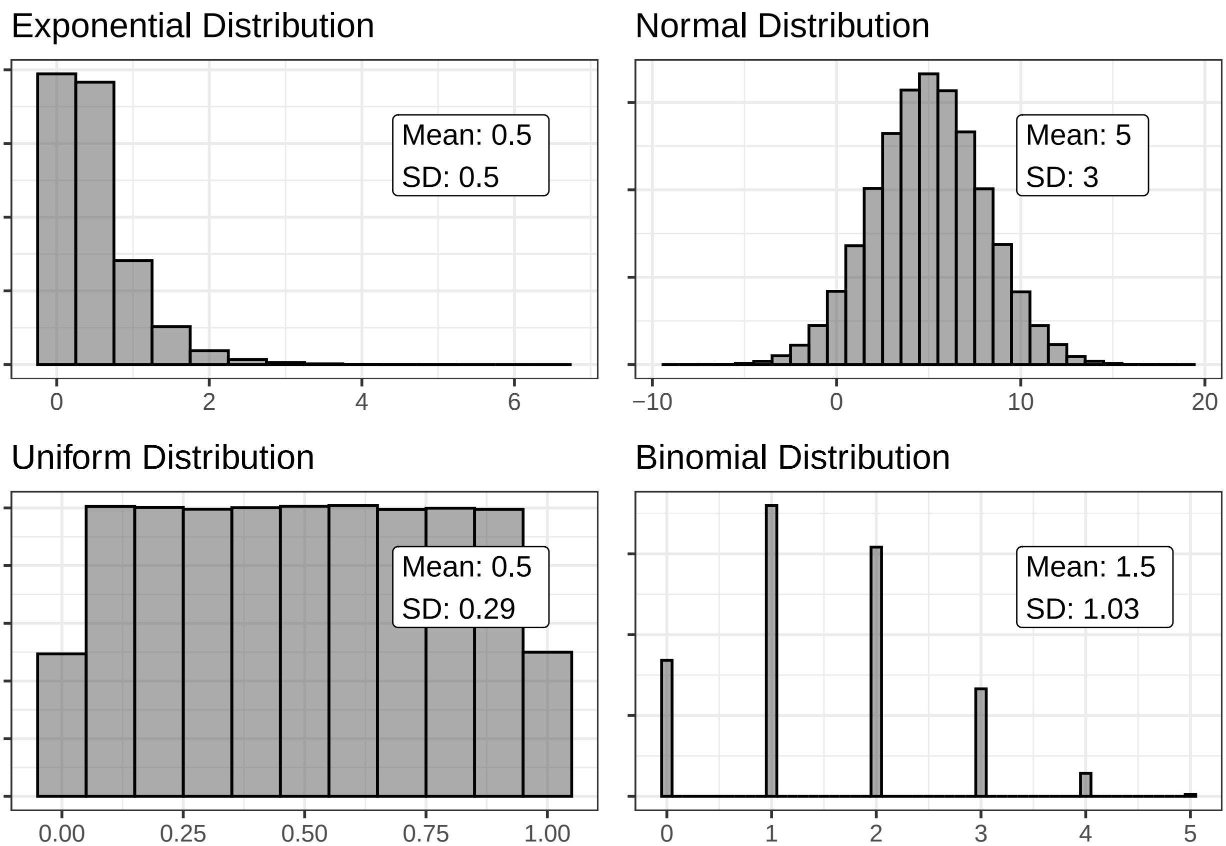
We now have:

- Pop. 1 $\sim$ Unif[0, 1]
- Pop. 2 $\sim$ Exp($\lambda = 2$)
- Pop. 3 $\sim$ N($\mu = 5$, $\sigma^2 = 9$)
- Pop. 4 $\sim$ Bin($n = 5$, $p = 0.3$)

Each of these distributions has a different mean, variance, shape, and support.

The plots below show each simulated population distribution. Recall the mean and variance formulas for each:

- Uniform: $E[X] = \frac{a+b}{2} = 1/2$, $Var(X) = \frac{(b-a)^2}{12} = 0.08$
- Exponential: $E[X] = 1/\lambda = 1/2$, $Var(X) = 1/\lambda^2 = 0.25$
- Normal: $E[X] = 5$, $Var(X) = 9$
- Exponential: $E[X] = np = 1.5$, $Var(X) = np(1-p) = 1.05$

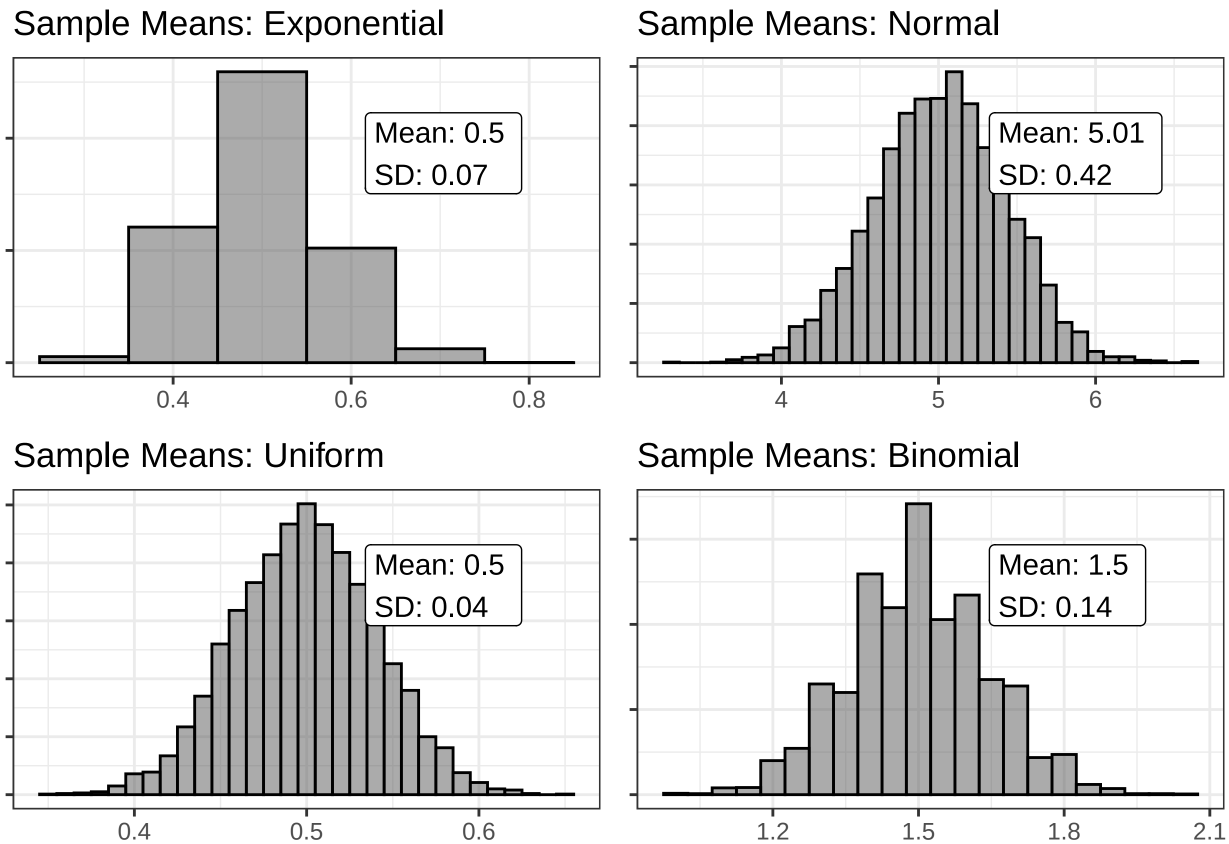


Now let's take sample draws from these distributions. For each sample, compute the average. Repeat this 5,000 times, each with a sample size of 50.

I plot the sample mean distributions on the next page. (The code I use to draw from the populations and plot the sample means is at the end of this document)

Note how the sample means always follow a normal distribution. Also note that the mean and variances align with the formulas:

$$E[\bar{X}] = \mu \qquad Var(\bar{X}) = \frac{\sigma^2}{n}$$


Sample Means: Exponential
Mean: 0.5
SD: 0.07
0.4
0.6
0.8
Sample Means: Normal
Mean: 5.01
SD: 0.42
4
5
6
Sample Means: Uniform
Mean: 0.5
SD: 0.04
0.4
0.5
0.6
Sample Means: Binomial
Mean: 1.5
SD: 0.14
1.2
1.5
1.8
2.1

Code for plotting functions

```r
library(tidyverse)
#----------------------------------------------
# Plot functions
#----------------------------------------------
plot_pop <- function(pop_dist, binwidth, dist_name, ann_x=NULL, ann_y=NULL) {

  pop_mean <- round(mean(pop_dist), 2)
  pop_sd   <- round(sd(pop_dist), 2)
  pop_N    <- length(pop_dist)

  ann_x <- ifelse(is.null(ann_x), 0.65, ann_x)
  ann_y <- ifelse(is.null(ann_y), 0.7, ann_y)

  data.frame(val = pop_dist) |>
    ggplot(aes(x = val)) +
      geom_histogram(binwidth = binwidth, alpha = 0.5, color="black") +
      xlab(NULL) + ylab(NULL) +
      labs(title = paste0(dist_name, " Distribution")) +
      theme_bw() +
      scale_y_continuous(labels = NULL) +
      #scale_x_continuous(labels = NULL) +
      annotate("label", x = I(ann_x), y = I(ann_y),
               label = paste("Mean:", pop_mean,
                             "\nSD:", pop_sd),
               color = "black", hjust = "left")
}

plot_means <- function(pop_dist, num_samples, sample_size,
                       binwidth, dist_name, ann_x=NULL, ann_y=NULL) {

  mean_draws <- replicate(num_samples,
                          mean(sample(pop_dist, sample_size, replace = TRUE)))

  s_mean <- round(mean(mean_draws), 2)
  s_sd   <- round(sd(mean_draws), 2)

  ann_x <- ifelse(is.null(ann_x), 0.6, ann_x)
  ann_y <- ifelse(is.null(ann_y), 0.7, ann_y)

  data.frame(val = mean_draws) |>
    ggplot(aes(x = val)) +
    geom_histogram(binwidth = binwidth, alpha = 0.5, color="black") +
    xlab(NULL) + ylab(NULL) +
    labs(title = paste0("Sample Means: ", dist_name)) +
    theme_bw() +
    scale_y_continuous(labels = NULL) +
    annotate("label", x = I(ann_x), y = I(ann_y),
             label = paste("Mean:", s_mean,
                           "\nSD:", s_sd),
             color = "black", hjust = "left")
}
#-------------------------------------------------------------------------------
```

```r
#-----------------------------------------------------------------------------
```

```r
library(gridExtra)

# Plot population distributions
grid.arrange(
  plot_pop(population_exp, binwidth = 0.5, dist_name = "Exponential"),
  plot_pop(population_norm, binwidth = 1, dist_name = "Normal"),
  plot_pop(population_unif, binwidth = 0.1, dist_name = "Uniform"),
  plot_pop(population_bin, binwidth = 0.1, dist_name = "Binomial"),
  ncol = 2)

# Take repeated samples and plot the sample means
grid.arrange(
  plot_means(population_exp, num_samples = num_samples,
             sample_size = sample_size, binwidth = 0.1,
             dist_name = "Exponential"),
  plot_means(population_norm, num_samples = num_samples,
             sample_size = sample_size, binwidth = 0.1,
             dist_name = "Normal"),
  plot_means(population_unif, num_samples = num_samples,
             sample_size = sample_size, binwidth = 0.01,
             dist_name = "Uniform"),
  plot_means(population_bin, num_samples = num_samples,
             sample_size = sample_size, binwidth = 0.05,
             dist_name = "Binomial"),
  ncol = 2)
```